Integrated analysis of lithosphere-ionospheric coupling associated with the 2021 M_w 7.2 Haiti earthquake from multiple satellites

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APPROVAL BY BOARD OF EXAMINERS

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Dr. Munawar Shah

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ABSTRACT

Earthquake (EQ) induced possible ionospheric anomalies has provided significant information about upcoming main shock. This study presents the abnormal ionospheric perturbations associated with M_w 7.2 Haiti EQ on August 14, 2021 with geographical coordinates (18°N, 73°W) and shallow hypocentral depth of 10 km (6.2mi) from data of permanent Global Navigation Satellite System (GNSS) stations near the epicenter and then validation is done by using data of Swarm satellites. The ionospheric electron content from different satellite showed a significant anomalous pattern within a time window of 5-10 days prior to main shock. The significant positive anomalies started to begin on August 10, 2021 rendered over epicenter till. The epicenter is located at the merge of two tectonic plates and a complex fault system of three fault lines. Therefore, the evolution of gases from lithosphere due to stress rock initiated air ionization, followed by ionosphere perturbation within EQ preparation period under the hypothesis of Lithosphere Atmosphere Ionosphere Coupling (LAIC).

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List of Abbreviations

SATNAV	Satellite Navigation
GNSS	Global Navigation Satellite Systems
GPS	Global Positioning System
MEO	Medium Earth Orbit
GLONASS	Global'naya Navigatsionnaya Sputnikova Sistema
EU	European Union
BDS	BeiDou navigation satellite system
GEO	Geostationary Earth Orbit
IGSO	Inclined Geosynchronous Orbit
SBAS	Satellite based augmentation system
SAR	Search and Rescue
WGS	World Geodetic System
EQ	Earthquake
TEC	Total Electron Content
LST	Land Surface Temperature
LAIC	Lithosphere Atmosphere Ionosphere Coupling
USGS	United States Geological Survey
STEC	Slant TEC
VTEC	Vertical TEC
IQR	Inter-quartile Range
CWT	Continuous Wavelet Transformation

UV	Ultraviolet
EM	Electromagnetic
EUV	Extreme Ultraviolet
HF	High Frequency
UB	Upper Bound
TECU	Total Electron Content Unit
LB	Lower Bound
NN	Neural Network
MLP	Multi-Layer Perceptron
NARX	Nonlinear AutoRegressive eXogenous
MODIS	Moderate Resolution Imaging Spectroradiometer
UT	Universal Time
ML	Machine Learning
URSI	International Union of Radio Science
TIR	Thermal Infrared Radiation
ECSS	European Cooperation on Space Standardization
hmF2	F2-peak height
SIAs	Seismic Ionospheric Anomalies
NmF2	F2 layer peak electron density
STEC	Slant Total Electron Content
LT	Local Time

1 HAITI EARTH QUAKE 2021 BACK GROUND

1.1 Back Ground

The evaluation of strong earth quake Mw 7.2, having depth 10 km (6.2 mi) in Haiti, on 14th August 2021, at 1229 hours UTC. The epicenter of Haiti earth quake is located at Latitude 18.4340 ° N, and at Longitude 73.4820° W, with 2248 deaths reported and 12763 injured.

Date	Magnitude	Causalities
14th August 2021	7.2	2248
12th January 2010	7.0	300,000

Table. 1.1. Haiti Earthquakes and respective causalities reported (https://www.usgs.gov/).

1.2 Objective and Motivation

In order to study and analyze the effects of EQ energy waves on ionosphere and evaluate seismic ionospheric anomalies within 20 days prior and 10 days after earth quake of Haiti EQ over Caribbean Region for epicenter having four fault zones around and located between two tectonic plates, using receiver data from GNSS-VTEC and validation of results from atmospheric indices and SWARM data and further evaluation of seismo ionospheric anomalies Haiti Mw 7.2 earth quake using GNSS – VTEC and validation of results using atmospheric indices and data from three SWARM satellites before the main shock for complex fault system which is located around epicenter

Epicenter of Haiti EQ Mw 7.2 is on a fault line and at the merge of two tectonic plates around a complex fault system of three fault lines so in order to confirm both LAIC and initiation of Ionosphere perturbations due to release of gases from cavities of stressed rocks around location of epicenter, during seismic preparation zone for quiet geomagnetic storms and their propagation to ionosphere due to atmospheric ionization

1.3 Content of Study

This paper has been divided in six chapters

Chapter 1: Brief introduction of Haiti Earth Quake 2021 along with seismic events took place between 11^{th} to 15^{th} August 2021.

Chapter 2: Introduction

Chapter 3: Literature Review

Chapter 4: Methodology adopted

Chapter 5: Results of this study

Chapter 6: Precise conclusion of results

2 INTRODUCTION

2.1 Seismology monitoring via GNSS

It is a fact that EQ causes huge damage to human beings and infrastructure every year. There are lot of different reports about early warning system of the possible natural hazard from various ground and satellite observations [1], [2], [3], [4], [5]. Moreover GNSS, DEMETER and Swarm satellites also provide deep insights on seismic precursor [6], [7], [8], [9]. Moreover, the ionospheric data is studied with different statistical and mathematical methods to find a particular pattern in the form of positive or negative anomaly in the EQ preparation zone with in a specific time window [10], [11]. Many reports also correlates ionospheric anomalies associated with geomagnetic storm [11], [12] and a report denied the ionospheric EQs anomalies with the current satellite cluster [13]. The seismic energy along the fault lines are most probably more than the surrounding and the change in pattern of releasing radon gas or energy after EQ or seismic activity [10]. The variations in atmospheric and ionospheric layers over the epicenter are indication of release of seismic energy, which may follow by an EQ [24].

2.1.1 Seismic ionospheric anomalies

Seismic ionospheric anomalies over the epicenter may be described in three different manners: positive holes (p-holes) emissions[14] [15], emanation of Radon [16] and Acoustic Gravity Waves (AGWs) [17].

2.1.2 Positive holes Emission

Release of Positive holes (P-holes) from fault lines under stress in tectonic regions may trigger air ionization followed by the delays in the path of propagation of radio signals [14]. Air ionization may be due to the coupling of lithosphere-atmosphere-ionosphere coupling (LAIC) process [16], [17], [18]. The initiation of chemical process in the atmosphere triggered by the release of p-holes from EQ regions further produced ionospheric anomalies. The p-holes hypothesis proposed the Earth and ionosphere as two ends of a capacitor, where charges flow from one end to another to form ionosphere perturbations

2.1.3 Radon Emanation

EQ prone areas cause air ionization and atoms/molecules propagation to ionosphere for delays in radio signals propagation. It is likely possible during the preparation period of EQ seismo ionospheric anomalies showed positive perturbations as a result of release of radon in gaseous state from fault lines of tectonic stressed rocks. Radon gas forms naturally and releases from compressed rocks over epicentral area when radioactive metal such like radium, thorium or uranium disintegrate in fault line, may it be in any level of soil or in sub-surface water due to tectonic plates movement and causes variation and perturbation in ionosphere. Increase in concentration of radon gas from Earth's crust along the fault lines in the seismogenic regions specially perturbs the passage of radio signals propagation [19], [20], [21], [22], [23].

2.1.4 Acoustic Gravity Waves

After the investigation of the energy diffusion route "from below" in the lithosphere-atmosphere-ionosphere-magnetosphere system, Acoustic gravity waves (AGW) are found as the most likely energy transporters for troposphere-ionosphere connection. The atmosphere acts as AGW amplifier, amplification factors ranging from 10^3 (region E of ionosphere) to 10^5 (region F of ionosphere). At high elevations, air pressure oscillations induced by meteorological sources unavoidably became nonlinear, causing wave decay and dissipation [17].

2.2 Applications of GNSS in field of seismology

2.2.1 Monitoring the motion of tectonic plates in event of earth quake

The global network of IGS provides seismic activity information with reference from motion of tectonic plates. Differential GNSS and Precise Point Positioning techniques are being used to monitor changes after earthquakes [66].

2.2.2 Monitoring of Atmospheric and Acoustic gravity waves generation

Atmospheric gravity waves generate after tsunami events and also Acoustic gravity waves generates after earth quake and both causes modulation of electron or plasma density in ionosphere and this modulation can be measured by using Differential GNSS [66].

3 LITERATURE REVIEW

In all over the world, earth quakes cause huge damage to human life and infrastructure. Extensive Research is being carried out to provide early warning regarding potential hazards and to minimize the damage and provide maximum protection to humans [8]. There have been reports received from various satellite observations about earth quake induced anomalies in ionosphere before main shock and after earth quake [1]. Dobrovolsky equation $\rho = 10^{0.43M}$ km [24] has been used in calculation region [24]. Different statistical models had been applied to the ionospheric data and assigning a particular (positive or negative anomalies) perturbation to preparation zone which surrounds the epicenter of earth quake before and after earth quake [2].

3.1 LAIC Phenomenon in earth quake ionospheric anomalies

According to Freund [12], [13] due to release of energy in shape of positive charges from fault zone in between tectonic plates causes disruption in atmosphere and ionosphere respectively. Daneshvar and Freund [23] have indicated rise of P-Holes in atmosphere due to thermal uplift and these P-Holes may cause variation in total electron content (TEC) of ionosphere [8], [10] as eminent variations in Slant TEC or Vertical TEC values before seismic event is an indicator of potential earth quake. Pulinets [14] has explained the random gas emission by under stressed rocks. Before earth quake, release of energy from stressed rocks of fault zones and the further its propagation from lithosphere to atmosphere and ionosphere [4].

3.1.1 Gas emissions

Seisomo Ionospheric Anomalies in earth quake preparation zone over epicenter may be describe in three different manners such as, positive holes (p holes) emissions [12, [13] generation radon atoms (which is known as Radon emanation) and transmission of acoustic gravity waves due to radium decay from earth quake prone regions [14] much before earth quake. Air ionization due to rock and air interaction results in initiation of positive ions specially in earth quake preparation zones due to stress in rocks [13]. This stress in rocks results in initiation of positive ions and this activation leads to emission of P Holes [25], [26]. Freund [12] had explained about rocks comes under stress from tectonic force generated by either of adjacent tectonic plate and activates positive holes release to earth's exterior and further to atmosphere. Due to this process air ionization occurs and it cause disruption in radio signals and main reason of ions traveling upward is that surface of earth is negative and lower ionosphere is positive [25] [26]. Pulinets [14] describes that Radon emanation from earth quake prone areas causes initiation of main shock preparedness process. This process cause air ionization along with atoms propagation to ionosphere and result is the formation of seismo ionospheric anomalies [14].

3.1.2 Movement of tectonic plates

Sudden release of energy / radon gas by stressed rocks under fault zones at the boundary of tectonic plates in earth quake preparation zones [14]. The impending anomalies in ionosphere may also be caused by upward movement of radon gas along with sharp increase in concentration of radon gas in soil, in water or in air. The cracks in earth's crust along fault lines provides the passage for the release of radon gas or energy [25], [26].

Anomalies in earth quake may which may be result of energy released before seismic event [14] due to movement of tectonic plates near fault zones that had caused Lithosphere-Atmosphere-Ionosphere coupling phenomenon, as gases seeps out due to presence of fractures in rock structures. Emanating gas may be considered one of the causes in increasing pressure in Lithosphere and surface temperature may rise. seismic waves are generated due to emancipation of energy in the earth's lithosphere and this energy moves upwards via rupture in geological faults

[16], [20, [25], [26], [27].

3.2 Remote sensing of atmospheric anomalies

By analyzing the DEMETER data, reports have indicated earth quake perturbations over epicenter [7] and variations in slant total electron content (STEC), plasma density and temperature has been analyzed from data of SWARM satellites [27] and for evaluation of thermal anomalies in earth quake region, data from NASA's different satellites has been studied [8], [9], [25], [26], [27]. Thermal anomalies in atmosphere includes rise in air and surface temperature or enhancement in Outgoing Longwave Radiations due to escape of trapped gases from stressed rocks which came under pressure due to tectonic plates movement [13].

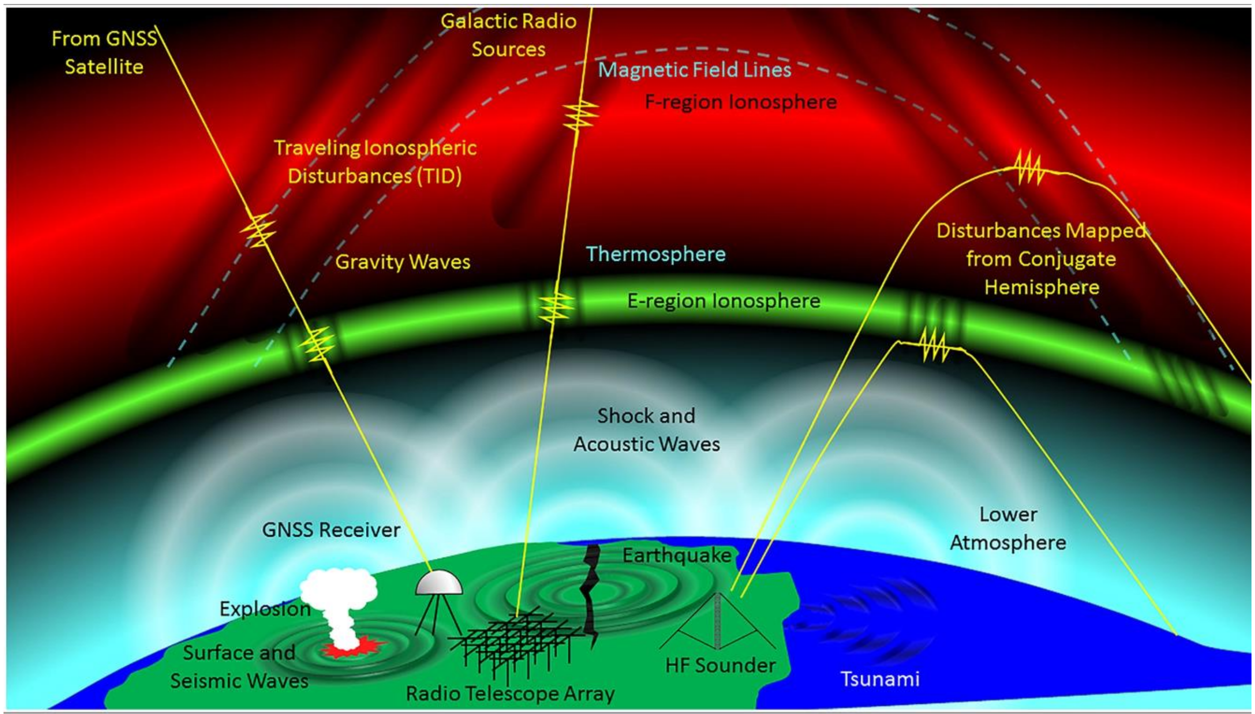

Fig.3.1. Schematic illustrating ionospheric response. Seismic wave is shown on left portion of figure. Earth quake is shown at middle portion. Tsunami in ocean at right. Shock wave and acoustic wave has been indicated as white semi circles. Perturbations shown by yellow zigzags. (https://agupubs.onlinelibrary.wiley.com/doi/full/10.1029/2017RG000594).

4 METHODOLOGY ADOPTED

The synchronized ionospheric and atmospheric anomalies in earth quake preparation zone over the epicenter have been analyzed from GNSS–VTEC and data from swarm satellites in domain of Slant Total Electron Content (STEC), plasma density and electron temperature (ET) [62] to establish a LAIC coupling hypothesis for the 2021 Haiti EQ on 14th August. A detailed study indicates a correlation exists between Vertical Total Electron Content of GNSS, Slant TEC data from swarm three satellites and other atmospheric indices over the epicenter within seismogenic zones to attribute the different characteristics of seismic anomalies [24], [25].

keeping focus on Vertical Total Electron Content (VTEC) data from IGS network in Dobrovolsky [24] region of Haiti EQ with the analysis of positive anomalies in ionosphere over epicenter [27], [28],[29]. Moreover, more emphasis has been on reliable and authentic statistical bound method for mean and standard deviation for finding ionospheric perturbations associated with the seismic event. Ionospheric data has been investigated for the August 14th, 2021, shallow hypocenter Haiti EQ of M_w 7.2.

4.1 Case Study of Area

The evaluation of abnormal ionospheric perturbations before strong Mw 7.2 of shallow hypocenter EQ, having depth 10 km (6.2 mi) in Haiti, on 14th August 2021, at 1229 hours UTC (0829 hours AM local time) when main shock was followed by 4 aftershocks and largest was of Mw 4.5 within 48 hours of main shock after main event. The geographical coordinates (Table. 4.1) of the epicenter are 18°N and 73°W (Fig.4.1).

EQ Date and Location	Latitude (°)	Longitude (°)	Depth (km)	Mw	Strike (°) (direction perpendicular to Dip direction)	Dip (°) (Angle with horizontal)	EQ Preparation zone (Radius) $R = 10^{0.43\,Mw}$

14th Aug'21, Nippes, Haiti at 1229 hours (UTC)	18.4340° N	73.4820° W	10	7.2	266°	51°	1247 km

Table 4.1: The detailed information about the Haiti EQ.

Haiti is located on the western part of Hispaniola Island in Caribbean Sea and Dominican Republic is on eastern side of Haiti. North American tectonic plate is on northern side of Haiti and Caribbean plate is on the southern side of Haiti and Gonave micro plate is located on western side of Haiti. Three fault zones surround Haiti; Septentrional Fault Zone is located on the northern side of Haiti, Oriento Fault Zone is located in north western side and Walton Fault Zone is located on western side of Haiti. Enriquillo-Plantain Garden Fault Zone passes through Haiti on which the epicenter of M_w 7.2 EQ is located. North eastern movements of about 0.5 m seismic slip on the Enriquillo-Plantain Garden Fault has been recorded after 14th August 2021 EQ. Haiti is located in the seismic prone region having strike slip faults and subduction faults (Fig.1) and at the boundary of two tectonic plates i.e. North American and Caribbean Plates. The detailed information about EQ is given in Table 1. Earth quake data has been noted down from website of United States Geological Survey (https://earthquake.usgs.gov/earthquakes/search).

Haiti EQ occurred on August 14, 2021 at Enriquillo-Plantain Garden Fault zone with hypocenter at shallow depth of 10 km and was measured strong MMI Level VIII (Modified Mercalli Intensity Scale 8th grade which indicates severe earth quake shaking https://en.wikipedia.org/wiki/Modified_Mercalli_intensity_scale). The total population of Haiti is 11.4 million and about 2 million people were exposed to this EQ and 2248 deaths were reported with 12763 injuries. Meanwhile, a total of 650,000 people required humanitarian assistance after August 14, 2021 (https://reliefweb.int/report/haiti/2021-haiti-earthquake-situation-report-2 september-10-2021). There are different measurements for the analysis of different Earth observation applications [31], [32], [33], [34], [35].

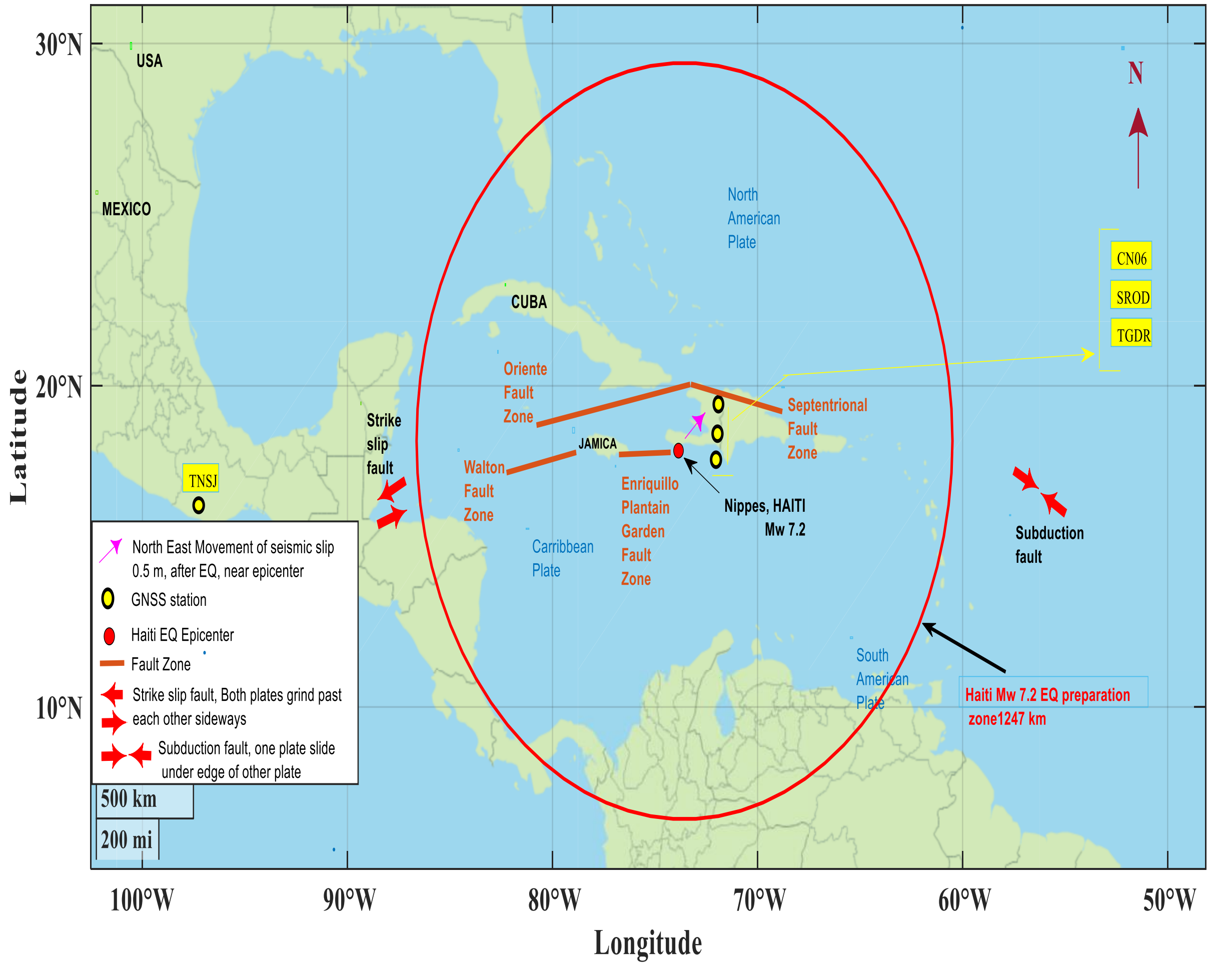

Fig.4.1. The geographical location of M_w 7.2 EQ with the yellow and black dots represent the location of GNSS stations. Red color circle represents the preparation zone estimated by Dobrovolsky [24] equation. The brown lines show the location of fault zones (information extracted from public domain data of USGS[60]) and modified it in Matlab (Version 2019a)-(www.mathworks.com). Red arrow indicates the north eastern movements (about 0.5 m) of seismic slip on the Enriquillo-Plantain Garden Fault after 14th August 2021 EQ, along with location of strike slip faults and subduction faults in EQ zone. All the information about USGS public domain data can be retrieved from https://www.usgs.gov/information-policies-and-instructions/copyrights-and-credits and https://earthquake.usgs.gov/earthquakes/eventpage/us6000f65h/dyfi/intensity [63].

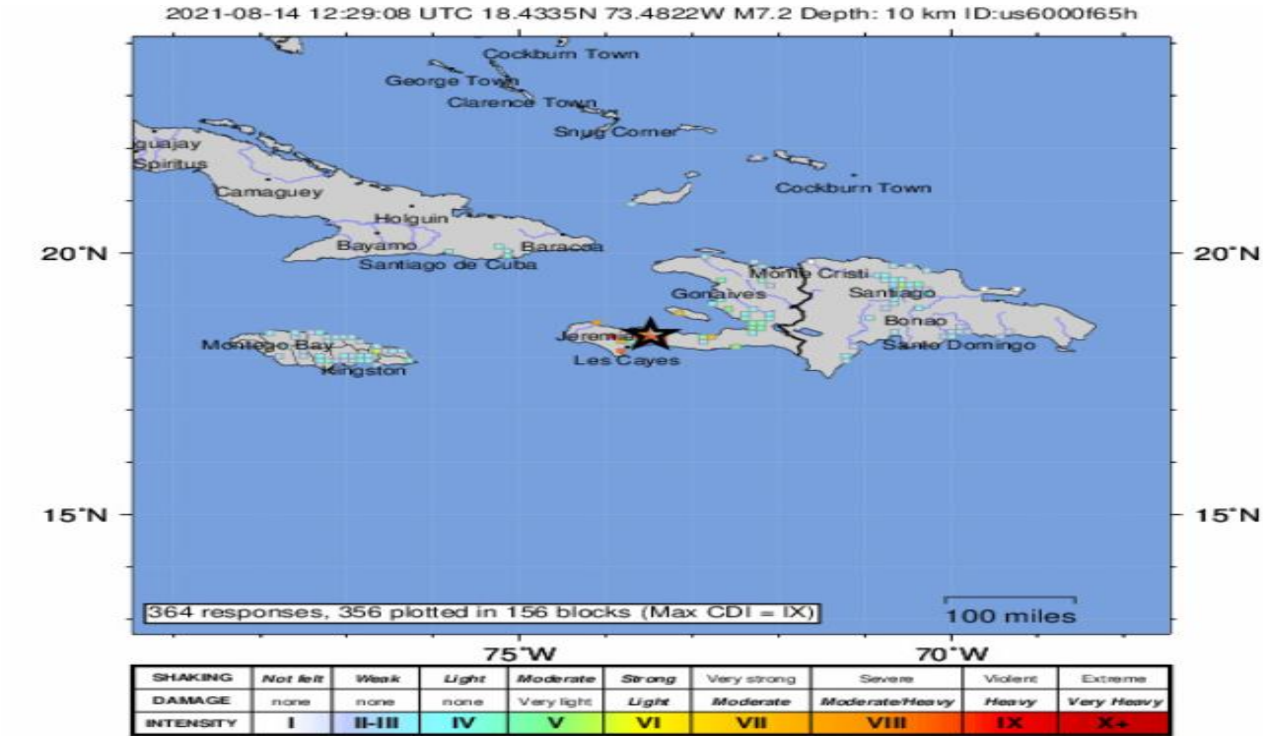

Fig.4.2. The USGS Shake Map of M_w 7.2, Haiti Earth Quake (14th August 2021), Source: https://earthquake.usgs.gov/earthquakes/eventpage/us6000f65h/dyfi/intensity.

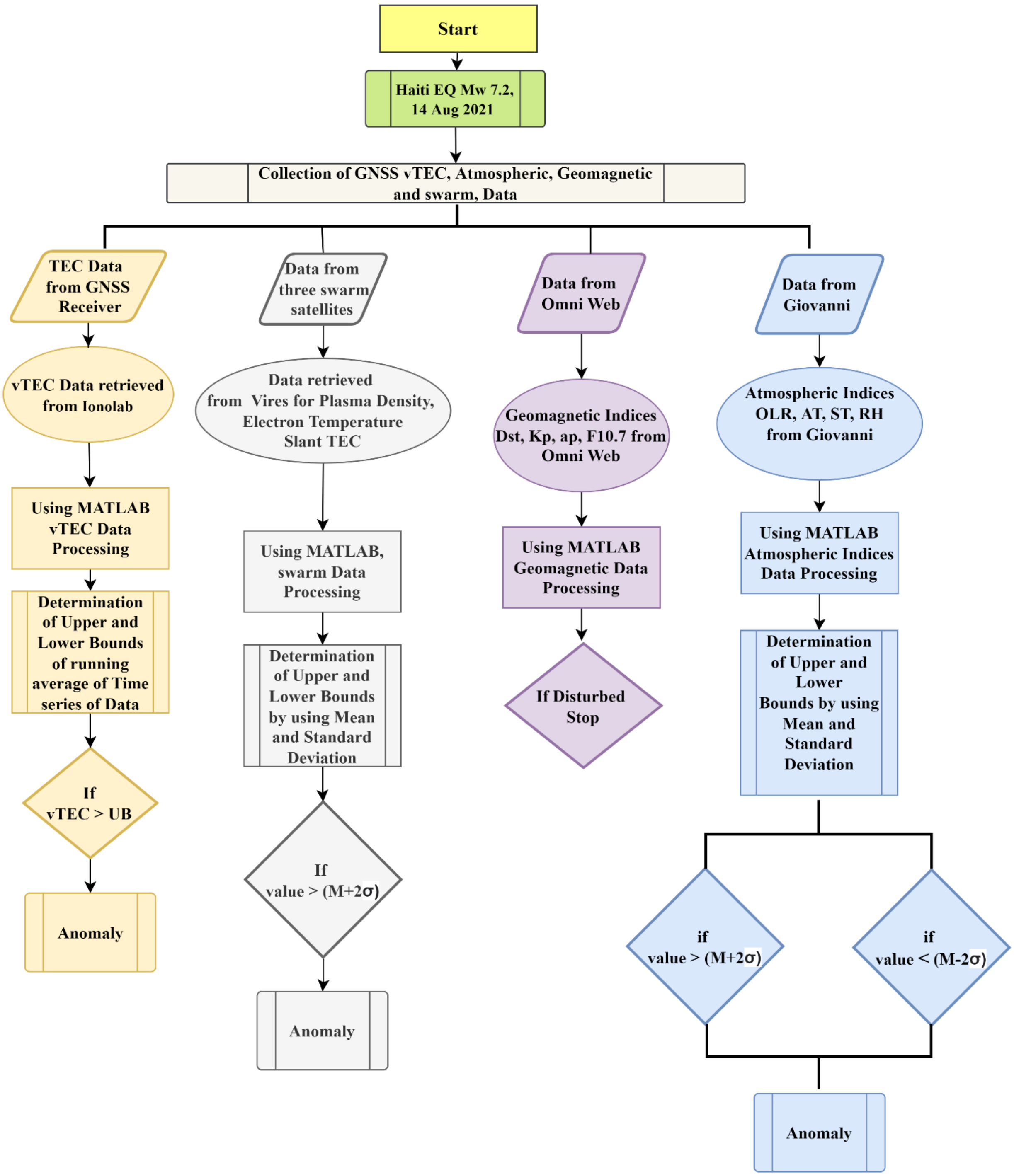

Table 4.2. The detailed research methodology flow chart showing step by step process followed from data collection, analysis, anomalies determination and deviation plotting.

4.2 GNSS-TEC retrieval

GNSS TEC data was collected for four IGS stations from IONOLAB http://www.ionolab.org/ [65], operating nearby the seismogenic zone to compute a relation between the EQ and ionosphere. Three IGS stations (CN06, SROD and TGDR) are located in Dominican Republic and one station (TNSJ) is located in Mexico (Table.5.1). By using the Dobrovolsky equation [24], the radius of the EQ preparation zone is approximately 1247 km.

$$R = 10^{0.43\,Mw} \qquad (1)$$

Where

R = Radius of EQ breeding area in km

Mw = Magnitude of EQ

In this study, the slant TEC astride the path of ray of the recorded signal from GNSS satellite to GNSS receiver is computed in unit of TECU (one TECU = 10^{16} electron /m^{2}). The thirty days TEC data from all the stations was collected from July 25, 2021 to August 24, 2021. Pre EQ-period is from July 25, 2021 to August 13, 2021 and from August 15, 2021 to August 24, 2021 is post EQ period. The main shock of EQ was on 14th August 2021, followed by four aftershocks within the next forty-eight hours. The largest after shock was of M_w 4.5 on August 15, 2021. The analysis of ionospheric TEC parameters in seismic zone of M_w 7.2 EQ shows anomalous TEC behavior on day-6, -5, -4 before main shock day in disturbed geomagnetic conditions (Kp>3: |Dst|<-30nT).

4.2.1 Conversion from Slant TEC (STEC) to Vertical TEC (VTEC)

Slant TEC has been retrieved alongside line of sight (LOS) from GNSS satellite to GNSS receiver, I is the ionospheric delay for signal L having frequency f and P is the Pseudo range of signal. As value of ionospheric delay is $I = \frac{40.28}{f^2}\, STEC$. In calculation of STEC two signals L_1 and L_2 having frequencies f_1 and f_2 were incorporated using the below equation.

$$STEC = \frac{f_1^2\, f_2^2}{40.28\left(f_2^2 - f_1^2\right)}\left(P - b_r - b_s - \epsilon_{arc} + \epsilon_l\right) \qquad (2)$$

Where I = Ionospheric delay for signal L having frequency f.

f = Frequency of signal L

P = Pseudo range of signal L

b_r = Differential code bias of receiver

b_s = Differential code bias of satellite

ϕ = Carrier phase measurement of signal L

ϵ_p = Noise effects on pseudo range measurements

ϵ_l = Multipath error

B_s = Inter frequency bias at satellite

B_r = Inter frequency bias at receiver

λ = Wavelength of signal L

N = Integer ambiguity of phase measurement

P_i = Smoothed pseudo range observation

$$P = P_1 - P_2 = I_1 - I_2 + b_s + b_r + \epsilon_p \quad (3)$$

$$\phi = \phi_1 + \phi_2 = I_2 - I_1 + \lambda_1 N_1 - \lambda_2 N_2 + B_s + B_r + \epsilon_l \quad (4)$$

$$P + \phi = \lambda_1 N_1 - \lambda_2 N_2 + B_s + B_r + \epsilon_l + b_s + b_r + \epsilon_p \quad (5)$$

Since, inter frequency bias B_s and B_r are considered as constants and value of ϵ_l is negligible, in pseudo range measurements as compared to ϵ_p, so average value in an incessant arc is taken as $(P + \phi)_{arc}$ in equation (5) [38], [39], [40], [41], [42].

$$(P + \phi)_{arc} = (\lambda_1 N_1 - \lambda_2 N_2)_{arc} + B_s + B_r + b_s + b_r + (\epsilon_p)_{arc} \quad (6)$$

$$P_i = (P + \phi)_{arc} - \phi \quad (7)$$

Subtracting equation (5) from equation (6), results in elimination of ambiguity terms

$$(P + \phi)_{arc} - (P + \phi) = b_s + b_r + (\epsilon_p)_{arc} - \epsilon_l$$

$$(P + \phi)_{arc} - \phi = P + b_s + b_r + (\epsilon_p)_{arc} - \epsilon_l \quad (8)$$

From equation (8), Smoothed pseudo range observation is equal to

$$P_i = P + b_s + b_r + (\epsilon_p)_{arc} - \epsilon_l \tag{9}$$

Pseudo range of signal L

$$P_i = I_1 - I_2 + b_s + b_r + (\epsilon_p)_{arc} \tag{10}$$

$$P_i - b_r - b_s - (\epsilon_p)_{arc} - \epsilon_l = (I_1 - I_2) \tag{11}$$

Substituting the f_1^2value of I in equation (11) as $I = \frac{40.28}{f^2}\ STEC$

$$P_i - b_r - b_s - (\epsilon_p)_{arc} - \epsilon_l = (\frac{40.28}{f_1^2}\ STEC - \frac{40.28}{f_2^2}\ STEC)$$

$$P_i - b_r - b_s - (\epsilon_p)_{arc} - \epsilon_l = (\frac{40.28}{f_1^2} - \frac{40.28}{f_2^2})STEC$$

$$STEC = \frac{f_1^2\ f_2^2}{40.28(f_1^2 - f_2^2)}\ (P_i - b_r - b_s - (\epsilon_p)_{arc} - \epsilon_l) \tag{12}$$

Considering $(\epsilon_p)_{arc} = \epsilon_{arc}$ and $P_i = P$ in equation (12)

$$STEC = \frac{f_1^2\ f_2^2}{40.28(f_2^2 - f_1^2)}(P - b_r - b_s - \epsilon_{arc} + \epsilon_l) \tag{13}$$

The VTEC may be calculated by using STEC (Schaer, S., 1999).

$$VTEC = STEC\ \left(\cos\left(sin^{-1}\left(\frac{R\ sinZ}{R+H}\right)\right)\right). \tag{14}$$

Where,

Z = Zenith angle of satellite

R = Radius of Earth

H = 350 km.

4.2.2 Anomalies determination process in Vertical TEC data

The time series of the GNSS stations are analyzed by the descriptive statistics method [43], [44], [45], [46], [47]. The abnormal TEC value is evaluated by a statistical analysis of running mean for 20 days before and 10 days after earth quake. The original TEC values beyond the upper bound of running mean is declared as an ionospheric anomaly. Moreover, Positive or Negative anomaly is the difference of original TEC from the upper bound of running mean.

Deviation TEC (dTEC) = $\text{TEC}_{\text{original}} - \text{TEC}_{\text{Upper Bound}}$

Ionospheric anomaly = $[\text{TEC}_{\text{original}} > \text{TEC}_{\text{Upper Bound}}]$

This criterion has been used to identify the earth quake related ionospheric anomalies.

4.3 SWARM data retrieval

Swarm three satellites is a mission launched by European Space Agency (ESA) for the observation regarding earth's magnetic field with high precision and accuracy. Swarm mission consists of three satellites designated as Alpha, Bravo, and Charlie (A, B and C) satellites. Alpha and Charlie are in low earth orbit at 460 km above mean sea level. Bravo satellite is also in low earth orbit at 510 km above mean sea. These orbital configurations are chosen largely to detect the variations in Earth's magnetic field [36]. It has many different sensors mounted on the satellites to observe the different atmospheric and electromagnetic variations and earthquakes of greater than seven magnitude worldwide [37]. The data for Haiti earth quake has been transferred via VirES (https://vires.services/). The thirty days data regarding Electron / Plasma Density (PD), Plasma / Electron Temperature (ET) and Slant Total Electron Content (STEC) from A, B and C satellites was noted down from 25th July 2021 to 24th August 2021 and in this regard pre-earth quake period is from 25th July, 2021 to 13th August, 2021 and post-earth quake period from 15th August 2021 to 24th August 2021. The main shock day of Haiti earth quake was on 14th August 2021.

4.3.1 Ionospheric anomalies determination process for three SWARM satellites

The ionospheric anomalies in the data of Swarm satellites for P Electron / Plasma Density (PD), Electron / Plasma Temperature (ET) and Slant Total Electron Content (STEC) data are detected by implementing the bounds of statistical mean and further calculated standard deviation (M ± 2σ). The original daily values of Swarm satellites data are sandwiched in the bound of mean and standard deviation, where the values beyond and below the bounds are denoted as anomalies. Deviation from upper or lower bound limits in the data plotted are ionospheric anomalies as detected by implementing the bound limits for mean (M) and standard deviation (σ),

Upper bound limit = $M + 2\sigma$ (1)

Lower bound limit = $M - 2\sigma$ (2)

Swarm A, B, C satellites data ionospheric anomaly calculation:

Ionospheric anomaly = Deviation = (Daily values – Upper bound)

Anomaly (PD, ET, STEC) = [swarm ionospheric daily $_{\text{original values}}$ > upper bound limit]

4.4 Atmospheric data retrieval

In order to obtain the parameters for atmospheric temporal data regarding Haiti earth quake 2021, indices regarding Relative Humidity (RH), Air Temperature (AT), Surface Temperature (ST) and Outgoing Longwave Radiation (OLR), the online database from NASA website, GIOVANNI https://giovanni.gsfc.nasa.gov/giovanni/ [61] has been used for acquisition of data for atmospheric indices and all these atmospheric values observed at 400 millibar pressure at an altitude of 23574 ft. The thirty days data was collected from July 25, 2021 to August 24, 2021, in this duration the pre-earth quake period is from July 25, 2021 to August 13, 2021 and post-earth quake period is from 15th August 2021 to 24th August 2021. The main shock of earth quake was on 14th August 2021. Analysis of atmospheric indices, indicates propagation of the LIAC phenomenon over seismic preparation zone from lithosphere to ionosphere layers. The original daily values of Relative Humidity (RH), Air Temperature (AT), Surface Temperature (ST) and Outgoing Longwave Radiation (OLR) data are sandwiched in the bound limits of mean and standard deviation, where the values beyond and below the bound limits has been denoted as anomalies. Deviation from upper or lower bound limits in the data plotted are atmospheric anomalies as detected by implementing the bound limits for mean (M) and standard deviation (σ), as mentioned in equation (1) and (2) respectively.

For atmospheric anomalies the bounds limits are considered as:

Upper bound limit = M + 2σ (1)

Lower bound limit = M - 2σ (2)

Atmospheric anomaly = Deviation = (Daily values – Upper bound)

Atmospheric anomaly = $[Atmospheric_{\text{ original}} > Atmospheric_{\text{Upper Bound}}]$

This criterion has been used to identify the earth quake related atmospheric anomalies.

4.5 Geomagnetic data retrieval

Geomagnetic data for solar and geomagnetic storm indices of Disturbance Storm Time Index (Dst), Indicator of global geomagnetic disturbance of Planetary K index (Kp) and (ap) has been retrieved from NASA OMNI web. Measure of the noise level generated by the sun at a wavelength of 10.7 cm at the earth's orbit (F 10.7) has been noted down for storm condition from https://omniweb.gsfc.nasa.gov and from web site of International Services of Geomagnetic Indices (ISGI), https://isgi.unistra.fr/index.php [64] as the geomagnetic storm indices are important indicators in determination of Lithosphere Atmosphere Ionosphere Coupling process. Thirty days data was collected, starting from July 25, 2021 to August 24, 2021, in this duration the pre-earth quake period is from July 25, 2021 to August 13, 2021 and post-earth quake period is from August 15, 2021 to August 24, 2021. The main shock of earth quake was on 14th August 2021.

Ionosphere scintillations mostly occur due to geomagnetic storms, so it is necessary to evaluate whether the under consideration seismic event occur in quiet or disturb storm condition [30].

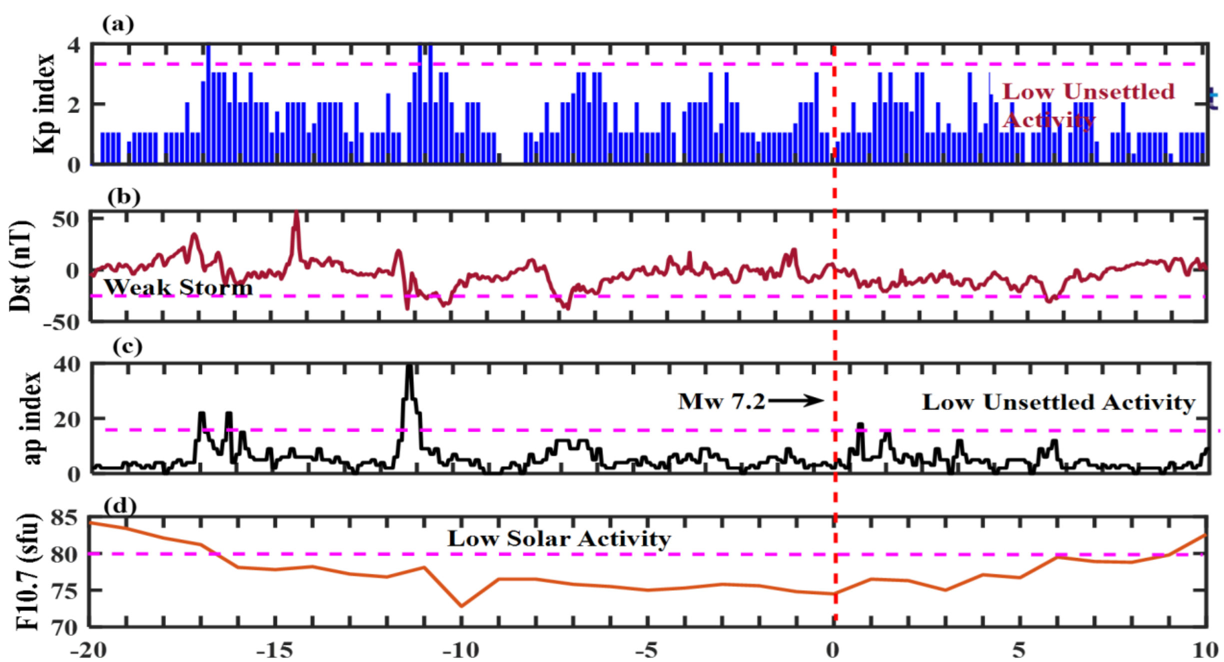

Fig.4.3. Panels representing geomagnetic storm indices of Dst, Kp, ap and F 10.7, Red color broken vertical line is for EQ day (data extracted from NASA/Giovanni).

5 RESULTS AND DISCUSSION

5.1 Results of analyses

Atmospheric data and ionospheric data disparity have been analyzed for seismogenic area of M_w 7.2 Haiti earth quake (on 14th August 2021) of shallow hypocenter. The radius of the preparation zone is 1247 km, which has been derived from Dobrovolsky equation ($R=10^{0.43M_w}$) [24]. Detailed geographical information of earth quake has been presented in table 4.1, earth quake occurred during disturbed geomagnetic activity with Kp > 3 and Dst < - 30nT, in Haiti region. Ionospheric variations and atmospheric indices values were evaluated from 25th July 2021 to 24th August 2021 (within 20 days before earth quake and 10 days after the earth quake day of Haiti earth quake 2021). Results of ionospheric and atmospheric anomalies are as follow.

5.1.1 Atmospheric indices anomalies

Atmospheric indices values of daily Outgoing Longwave Radiation (OLR), Relative Humidity (RH), Air Temperature (AT) and Surface Temperature (ST) were evaluated within 20 days before earth quake and 10 days after earth quake day of Haiti earth quake 2021. While statistically analyzing atmospheric indices in time series data over earth quake breeding zone, the deviation from upper and lower bounds of statistical mean and calculated standard deviation were considered anomalies.

5.1.2 Air Temperature and Relative Humidity anomalies

Maximum deviation from upper bound of Air Temperature time series values has been observed on day -5, with maximum value of 259 K (Fig.5.1), and in case of Surface Temperature, maximum deviation from lower bound of Relative Humidity time series values has been observed on day -5 with minimum value of 29% before main shock of Haiti EQ 2021.

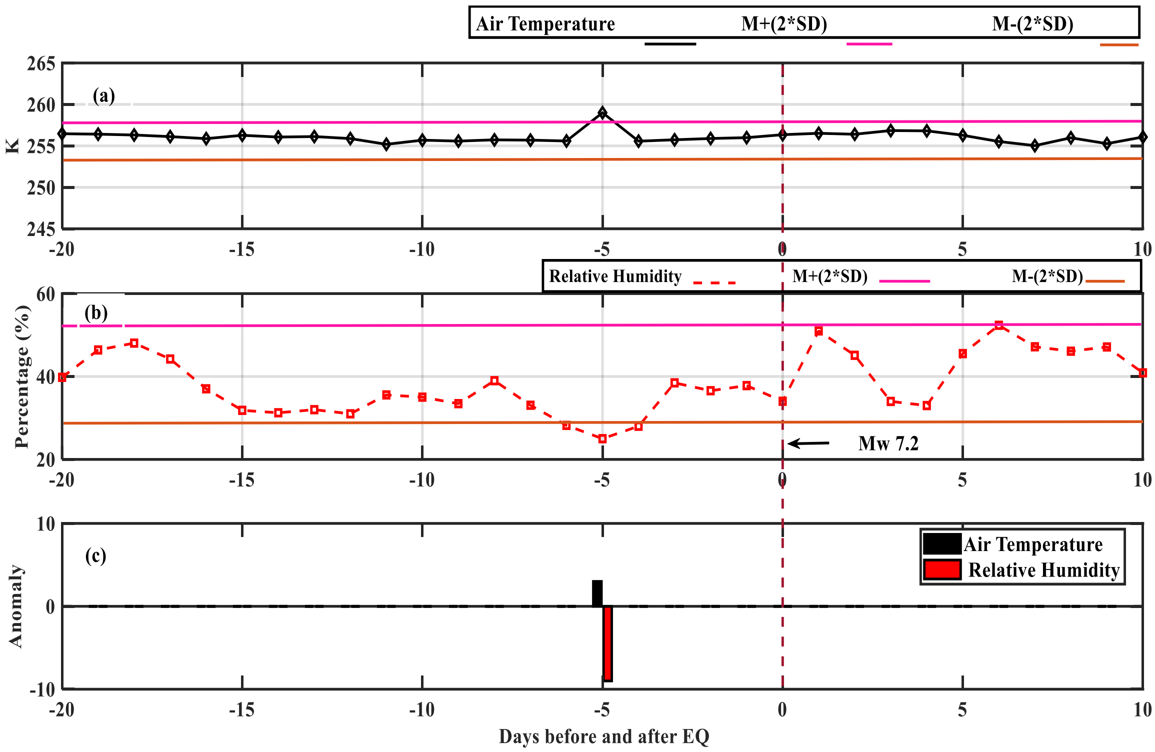

Fig.5.1. Graphical representation of indices of Air Temperature and Relative Humidity time series evaluation, (a) Air Temperature time series, (b) Relative Humidity time series over the epicentral zone with upper and lower statistical bounds of mean and calculated standard deviation, (c) variation of AT time series values, ST time series values, from upper and lower statistical bounds of mean and calculated standard deviation, are considered as anomalies. Red broken vertical line indicates the earth quake day.

5.1.3 Outgoing Longwave Radiation and Surface Temperature anomalies

The significant abnormality has been detected on 11th August 2021, while evaluating data of daily Outgoing Longwave Radiation (OLR) values at 400mb from 25th July to 24th August 2021, (Fig 5.1) the results plotted have shown a sudden and sharp rise in OLR values as observed on day -3 with maximum value recorded was 295 W/m^2 and then afterwards decline in OLR values was observed. While evaluating the data of Surface Temperature (ST) from 25th July to 24th August 2021 (Fig 5.1), rise in value of (ST) has been observed on day -4 before main shock day, with maximum value of 306 K was observed on 10th August 2021, then afterwards values remains within bounds limits.

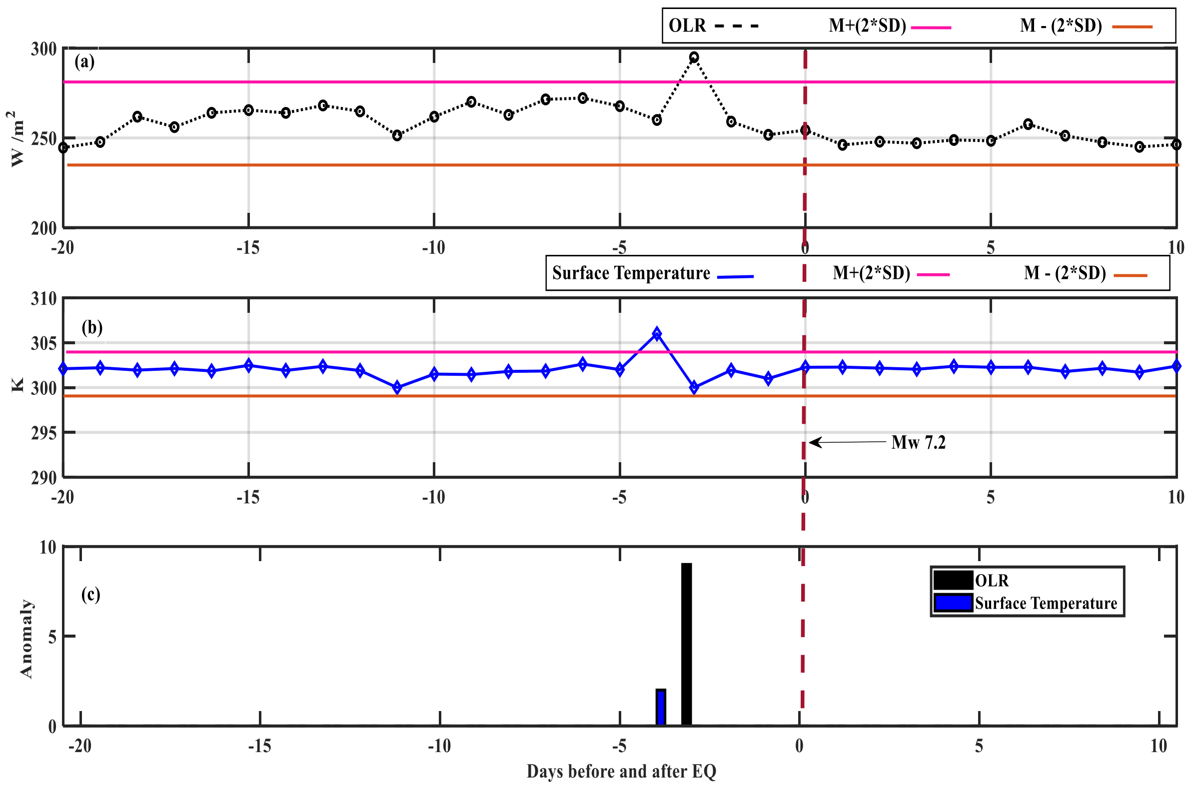

Fig.5.2. Outgoing Longwave Radiation and Surface Temperature time series evaluation (a) Outgoing Longwave Radiation time series, (b) Surface Temperature time series for Haiti earth quake breeding zone with upper and lower statistical bounds of mean and calculated standard deviation, (c) variation of OLR time series values and ST time series values, from upper and lower statistical bounds of mean and calculated standard deviation, are anomalies. The red broken vertical line indicates the EQ day.

5.2 Ionospheric Vertical Total Electron Content (VTEC) Anomalies

The ionospheric Vertical Total Electron Content (VTEC) anomalies were observed on day -6, -5, -4, when TEC data regarding four International GNSS Stations CN06, SROD, TGDR and TNSJ (Fig.4.1) was analyzed from 25th July 2021 to 24th August 2021, which is 20 days before Haiti earth quake and 10 days after earth quake day. IGS network station CN06, SROD and TGDR (Table. 5.1) showed clear VTEC anomalies on day -6, -5, -4, before the main shock, day as these stations are located inside the seismic preparation zone and the geographical location of CN06, SROD and TGDR stations is in Dominican Republic which is a neighboring country of Haiti (Fig.

4.1). IGS TNSJ which is located in Mexico (Fig.4.1), indicated anomalies on small scale as compare to rest of three stations so it means that earth quake energy was so huge that it also disturbed the ionosphere even far from the epicenter at about 2454 km (Table 5.1), as TNSJ station located outside the seismic preparation zone. VTEC of all ground GNSS stations except TNSJ (Fig.4.1) indicate the abnormal rise in original VTEC values in 6-day time window, before earth quake day. This deviation of original VTEC values beyond the upper bound indicates ionospheric anomaly in VTEC data as far as analyses of statistical bound method is concerned. The deviation is about 9–18 TECU during day time between UT 0800 hours to UT 1200 hours in the time window of 6 days before earth quake day. All GNSS stations indicates positive ionospheric anomaly (anomalous beyond upper bound).

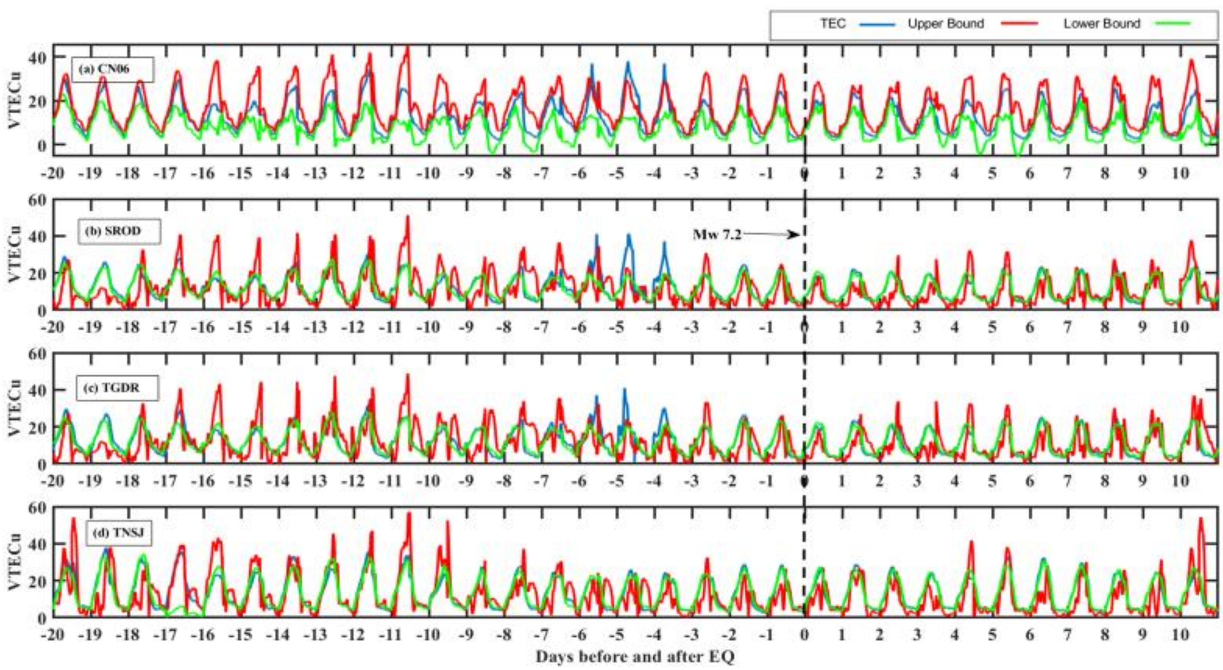

Fig.5.3. The original VTEC values noted at four IGS network (CN06, SROD, TGDR, TNSJ). Panel (a) exhibits the VTEC disparity at CN06 station for 31 days (20 days before earth quake and 10 days after earth quake day). Panel (b) shows the VTEC disparity at SROD station for 31 days (20 days before and 10 days after the EQ). Panel (c) shows the VTEC disparity at TGDR station for 31 days (20 days before and 10 days after the EQ). Panel (d) shows the VTEC disparity at TNSJ station for 31 days (20 days before and 10 days after the EQ) as this station is located in Mexico, outside the Dobrowolsky region so it shows very minute VTEC disparity. Blue line depicts VTEC values, while red lines depicts upper statistical bound and green line depicts lower statistical bound bounds. Names of GNSS stations on top left side of each graphical panel. The black broken vertical line represents the EQ day.

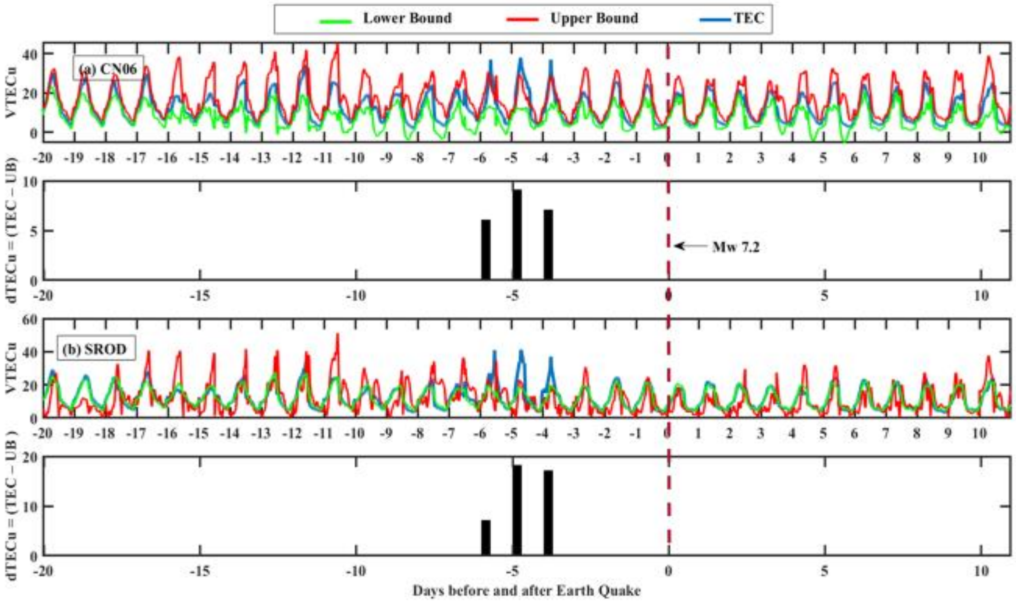

Fig. 5.4. Deviation vTEC = original vTEC – TEC $_{\text{Upper Bound}}$, the values of original TEC are exceeding from Upper Bound on Day (-6, -5, -4), for CN06 and SROD stations, before Haiti earth quake of Mw 7.2 EQ. Red broken vertical line represents the earth quake day.

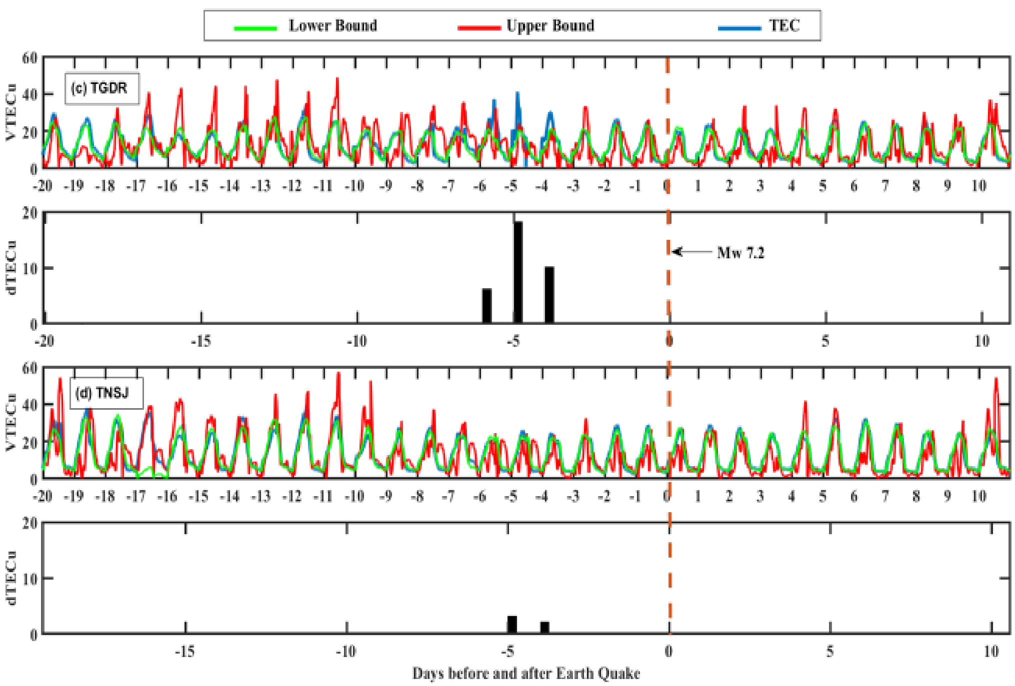

Fig. 5.5. Deviation vTEC = original vTEC – TEC $_{\text{Upper Bound}}$, the values of original TEC are exceeding from Upper statistical Bound on Day (-6, -5, -4), for TGDR and TNSJ stations, before Haiti earth quake of Mw 7.2. Red broken vertical line represents the EQ day.

S. no	GNSS Station	Distance from epicenter (km)	**Geographic Coordinates**		**Anomalous days within EQ period**
			Latitude (°)	Longitude (°)	Pre-seismic days
1.	CN06	300	18.7900° N	70.6560° W	-6, -5, -4
2.	SROD	253	19.4750° N	71.3410° W	-6, -5, -4
3.	TGDR	254	18.2080° N	71.0919° W	-6, -5, -4
4.	TNSJ	2454	16.1724° N	96.4895° W	-5 (Very less value on Day -5)

Table. 5.1. The detailed information about the corresponding IGS network and their respective distances from epicenter.

5.3 SWARM A, B, C satellites deviations and anomalies

Three satellites data was noted down which includes Slant Total Electron Content (STEC) and plasma density (PD) from 25th July 2021 to 24th August 2021, This period originates from 20 days before earth quake and also includes 10 days after earth quake day. Data was analyzed in order to verify the ionospheric data deviations and data from swarm satellites confirmed the deviations which has been indicated as anomalies ionospheric data. The plasma density and STEC variations from Swarm satellites also endorse deviations in VTEC as noted down from IGS network regarding Haiti earth quake. (Fig.14).

The analyses of Plasma density from Swarm three satellites (Fig. 8, 10, 12) clearly showing the variations on August 10, 2021 before earth quake day. The daytime enhancement in GNSS TEC data and in Plasma density data along with Slant TEC data from three Swarm satellites is clearly visible. It also shows that ionospheric anomalies began to initiate on August 10, 2021. Moreover, the rise in daytime ionosphere reached to maximum on August 10, 2021 (four days before the main shock).

5.3.1 SWARM A, C satellites Plasma Density and Slant TEC deviations

The data of swarm A and C, satellites indicate day time enhancement in values of Slant Total Electron Content (STEC) and plasma density (PD) in 6-day time window due to upward rise of gases over epicenter. For determination of anomalies the data of STEC and PD values of swarm A and C, satellites were analyzed from 25th July 2021 to 24th August 2021, which is 20 days before earth quake and 10 days after earth quake day of Haiti earth quake 2021. The values of plasma density (PD) and Slant Total Electron Content (STEC) from Swarm A and C satellites enhances beyond upper bound limit on day -5, this deviation indicates anomaly and plotted values exhibits variation within 6 days before the main shock day (Fig. 5.7 and 5.9, second and fourth panel). These variations are earth quake induced abnormal variations, as they differ from the rest of the measurements of PD and STEC.

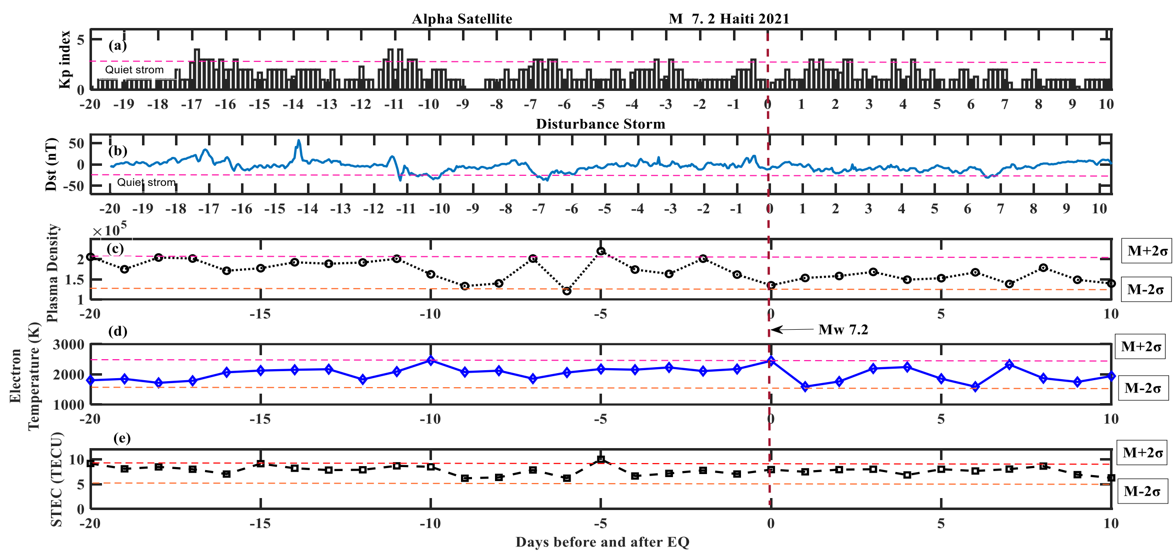

Fig.5.6. Panel (a) represents Planetary Index and panel (b) represents Disturb storm time storm for Haiti earth quake M_w 7.2. Panel (c) represents Plasma Density (N_e) for period of 20 days before earth quake and 10 days after earth quake. Panel (d) represents Electron Temperature (ET). Panel (e) represents Slant Total Electron Content (STEC) of Swarm Alpha satellite for M_w7.2 (Haiti). Red broken vertical line represents the EQ day.

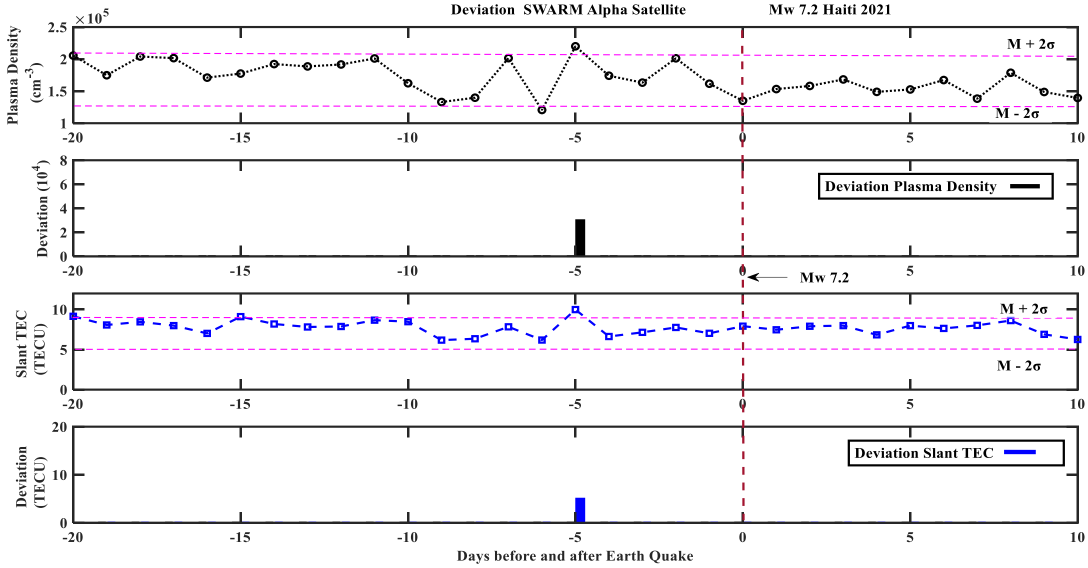

Fig.5.7. Panel (a) represents Plasma Density (N_e) data for period of 20 days before earth quake and 10 days after earth quake. Panel (b) represents variation of Plasma Density (N_e) from upper and lower statistical bounds of mean and calculated standard deviation as anomaly. Panel (c) represents Slant Total Electron Content (STEC) data for period of 20 days before earth quake and 10 days after earth quake. Panel (d) represents variation in STEC data from upper and lower statistical bounds of mean and calculated standard deviation as anomaly from Swarm Alpha satellite for M_w7.2 (Haiti). Red broken vertical line represents the earth quake day.

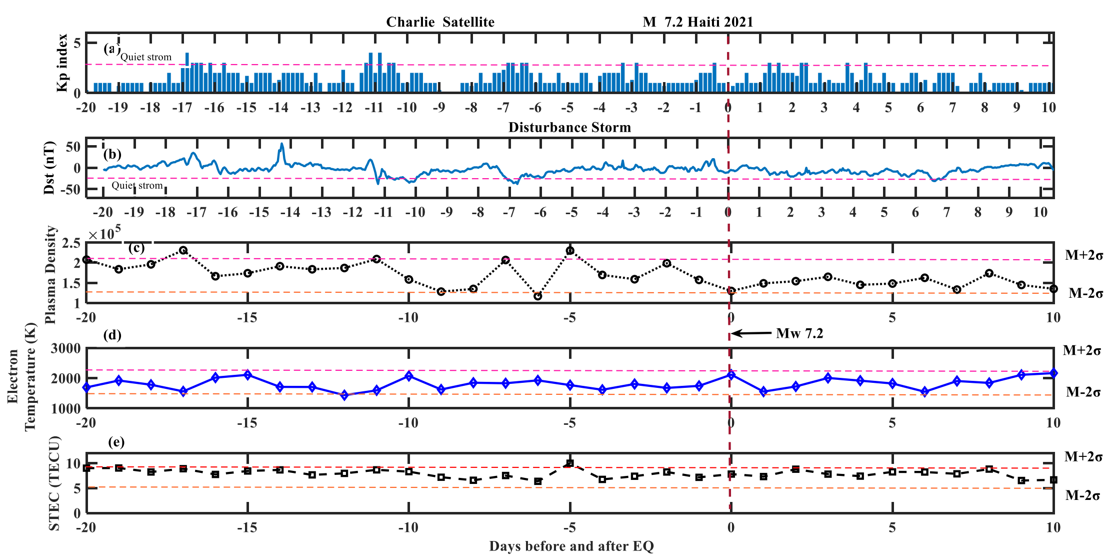

Fig. 5.8. Panel (a) represents Planetary Index and panel (b) represents Disturb storm time storm for Haiti earth quake M_w 7.2. Panel (c) represents Plasma Density (N_e) for period of 20 days before earth quake and 10 days after earth quake. Panel (d) represents Electron Temperature (ET). Panel (e) represents Slant Total Electron Content (STEC) of Swarm Charlie satellite for M_w7.2 (Haiti). Red broken vertical line represents the EQ day.

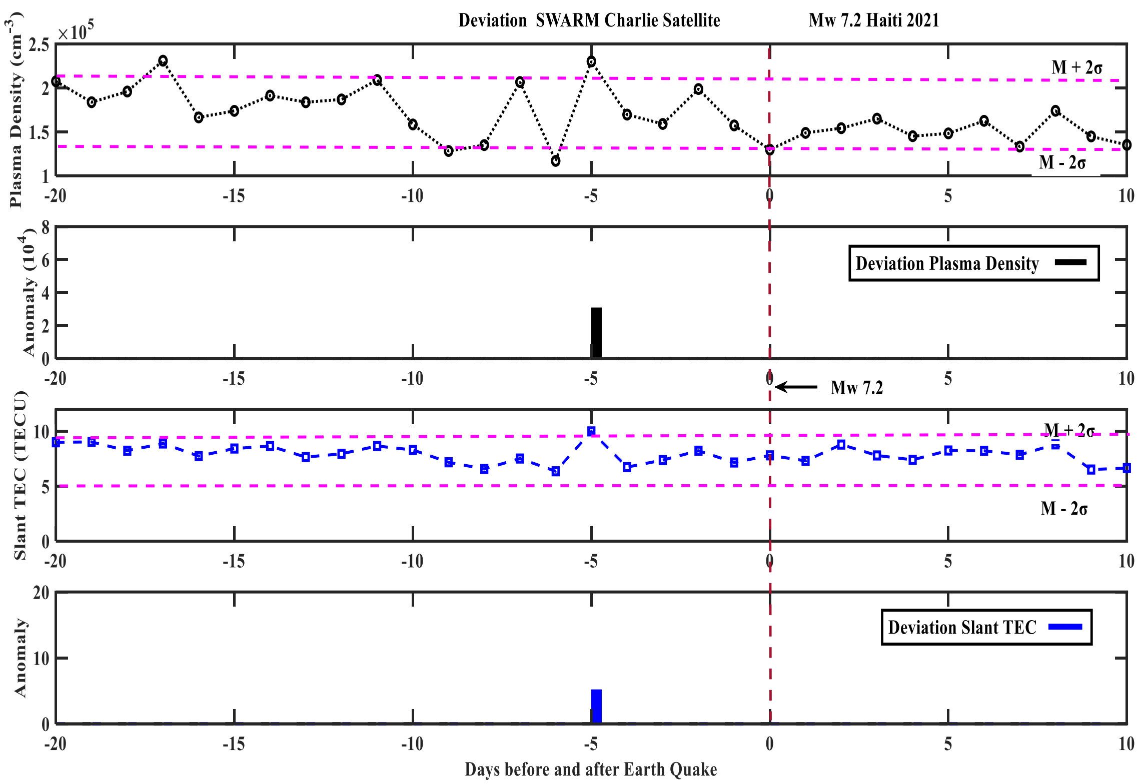

Fig.5.9. Panel (a) represents Plasma Density (N_e) data for period of 20 days before earth quake and 10 days after earth quake. Panel (b) represents variation of Plasma Density (N_e) from upper and lower statistical bounds of mean and calculated standard deviation as anomaly. Panel (c) represents Slant Total Electron Content (STEC) data for period of 20 days before earth quake and 10 days after earth quake. Panel (d) represents variation in STEC data from upper and lower statistical bounds of mean and calculated standard deviation as anomaly from Swarm Charlie satellite for M_w7.2 (Haiti). Red broken vertical line represents the earth quake day.

5.3.2 SWARM B satellite deviations and anomalies

The data of swarm B satellite do not any indicate day time enhancement in values of Slant Total Electron Content (STEC) and plasma density (PD) in 6-day time window. For determination of anomalies the data of STEC and PD values of swarm B satellite was analyzed from 25th July 2021 to 24th August 2021, which is 20 days before and 10 days after the main shock day of Haiti earth quake 2021. The values of plasma density (PD) and Slant Total Electron Content (STEC) from Swarm B satellite remained with in statistical bounds limits. (Fig. 5.11, second and fourth panel).

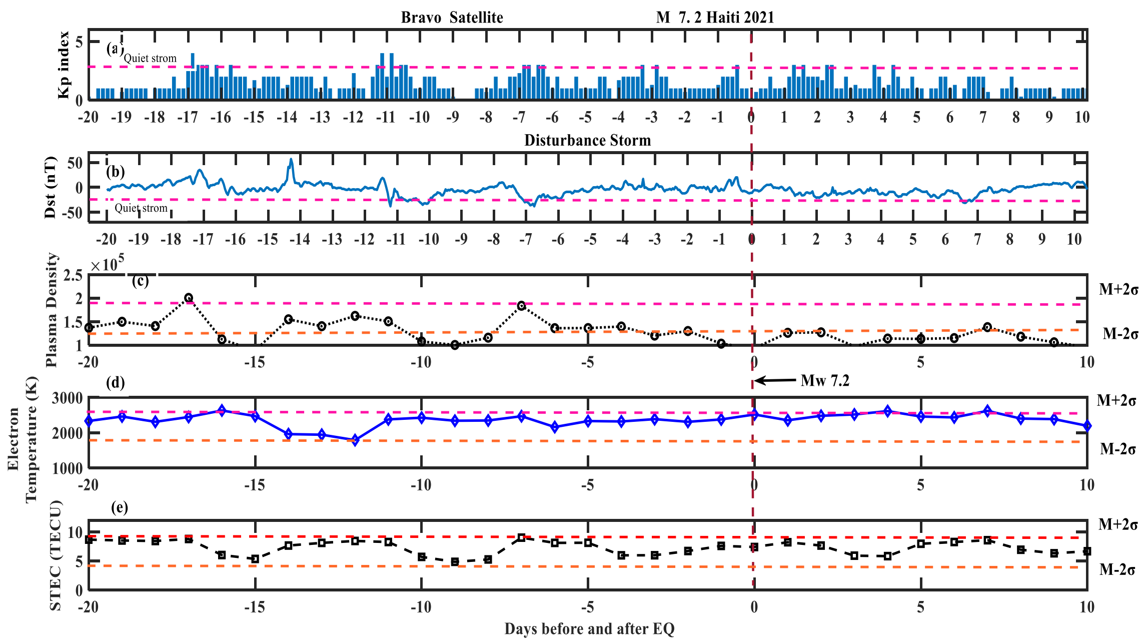

Fig. 5.10. Panel (a) represents Planetary Index and panel (b) represents Disturb storm time storm for Haiti earth quake M_w 7.2. Panel (c) represents Plasma Density (N_e) for period of 20 days before earth quake and 10 days after earth quake. Panel (d) represents Electron Temperature (ET). Panel (e) represents Slant Total Electron Content (STEC) of Swarm Bravo satellite for M_w7.2 (Haiti). Red broken vertical line represents the EQ day.

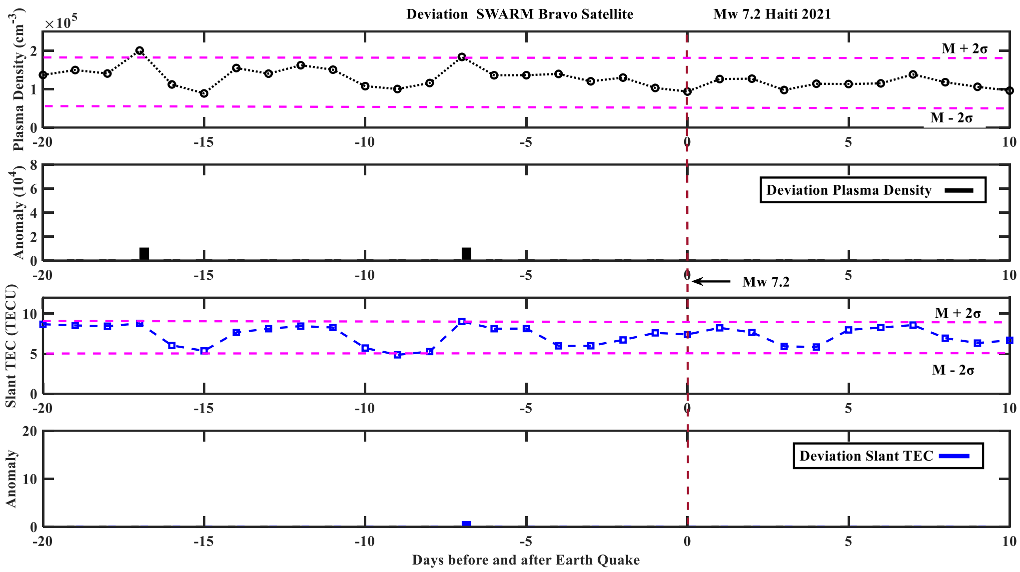

Fig.5.11. Panel (a) represents Plasma Density (N_e) data for period of 20 days before earth quake and 10 days after earth quake. Panel (b) represents variation of Plasma Density (N_e) from upper and lower statistical bounds of mean and calculated standard deviation as anomaly. Panel (c) represents Slant Total Electron Content (STEC) data for period of 20 days before earth quake and 10 days after earth quake. Panel (d) represents variation in STEC data from upper and lower statistical bounds of mean and calculated standard deviation as anomaly from Swarm Bravo satellite for M_w7.2 (Haiti). Red broken vertical line represents the earth quake day.

5.3.2 Comparison of daily values from Swarm A, B, C satellites of plasma density (N_e) and Electron Temperature (ET)

The variation in values of Plasma Density (PD) and Electron Temperature (ET) from swarm A and C satellites confirms enhancement in Plasma Density (PD). Maximum value of Plasma Density has been noted on day -5, in 6-day time window indicates anomaly. Values of PD and ET plotted for mutual analyses (Fig. 5.12. First and Third panel) and graphs exhibits maximum deviation on day -5 for Plasma Density from data of swarm A and C satellites. However, no significant deviation has been indicated by the data of PD and ET from swarm B satellite.

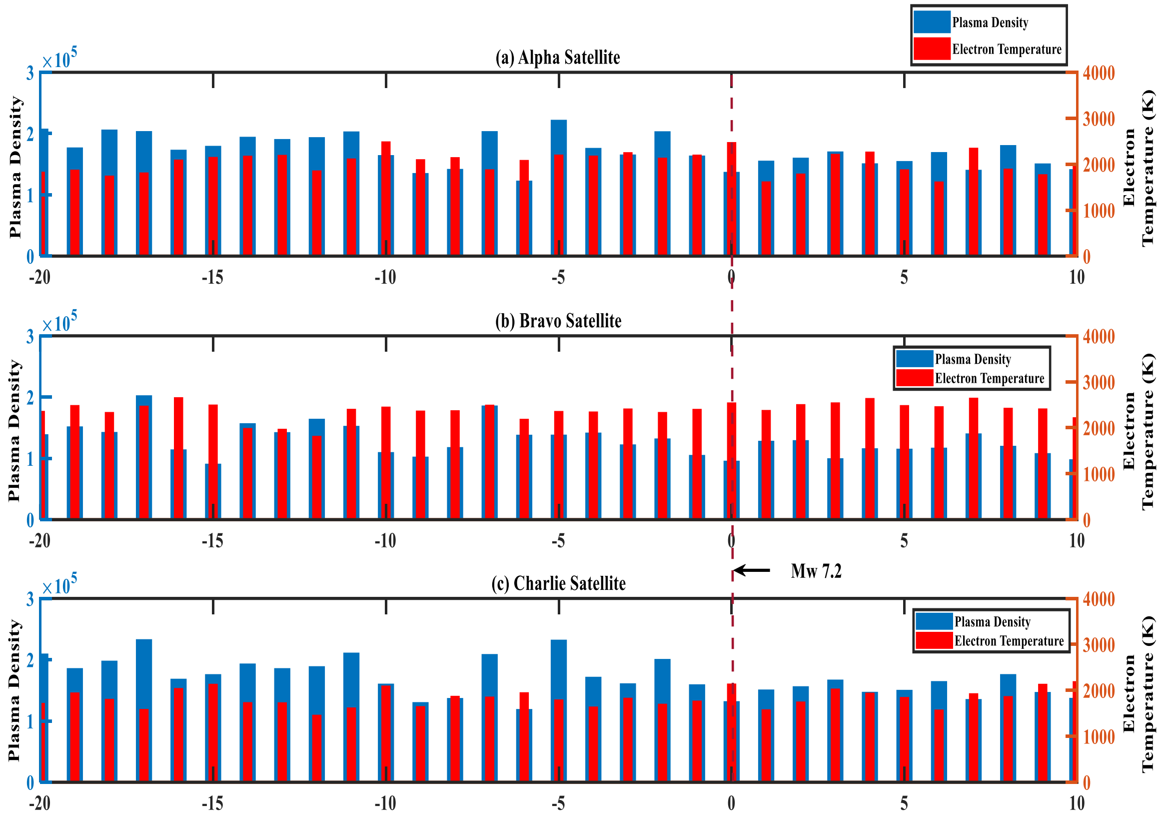

Fig.5.12. Panel (a) represents difference in values of Plasma Density (N_e), exhibits in red colored bar and Electron Temperature (ET), exhibits in blue colored bar for Swarm A satellite, Panel (b) represents difference in values of Plasma Density (N_e), exhibits in red colored bar and Electron Temperature (ET), exhibits in blue colored bar for Swarm B satellite, Panel (c) represents difference in values of Plasma Density (N_e), exhibits in red colored bar and Electron Temperature (ET), exhibits in blue colored bar for Swarm C satellite for Haiti earth quake of M_w 7.2. Red broken vertical line represents the EQ day.

5.4 Discussion on Results

The analyses of atmospheric and ionospheric indices have been done with an objective to evaluate phenomenon of coupling of Lithosphere with Atmosphere and Ionosphere in an area with complex fault zone and lies between two tectonic plates in order to further unfold the seismo-iono anomalies over seismic preparation zone. Examining the pre-seismic cursors in seismic preparation zone indicates atmospheric and ionospheric anomalies within 6-day time window before the main shock day, along with fact that these results are being supported by SWARM satellites data evaluation which also resulted in indication of anomalies. Summary of atmospheric, ionospheric and swarm anomalies observation with dates is shown in table 5.2

Evaluation of data has indicated an increase of 9 - 18 TECU in daily TEC from VTEC GNSS during day time in UT 0800-1200 hours before seismic event. The enhancement in ionosphere before the main shock confirms the emission of huge energy from seismogenic region due to tectonic stress building in rocks beneath epicenter, as OLR higher values are may be due to emission radon gas, values of OLR remained on higher side during EQ preparation period with maximum value on day -3, and Air temperature exceeding upper bounds limits on day -5.

These results (Table 5.2) are an evidence about the possible precursors of ionospheric anomalies associated with M_w 7.2 Haiti. Both the ground from IGS, and from swarm satellite observations indicates positive abnormal variations from the respective upper bounds [53], [54], [55], [56], [57].

Additionally, all the ground and space satellites confirm the abnormal ionospheric variations within 6 days with huge intensity. There is a possibility of abnormal amount of energy from the stressed rocks that can possibly perturbed the ionosphere during seismic activity. Moreover, this energy occurred in the form of abnormal Radon gas emission, and the abnormal ionospheric clouds can possibly move from the lithosphere to ionosphere due to p-hole hypothesis. Which is generated at hypocentral depth and further ionize the atmosphere over the seismogenic zone, following by excitation of ionospheric electron content during the seismic preparation period. After this electron content reached up to a certain threshold value, it causes to deviate the path of satellite signals during its journey towards the receiver stations [58], [59].

Anomaly Observation days with Parameters			
Parameter	**Exceeding Upper Bounds**	**M+2σ**	**M-2σ**
Ionospheric (VTEC) IGS CN06, SROD, TGDR	Day -6, Day -5, Day -4		
(VTEC) IGS TNSJ (Outside preparation zone, in Mexico)		Nil	
SWARM A (PD, STEC)		Day -5	
SWARM B (PD, STEC)		Nil	

SWARM C (PD, STEC)		Day -5	
Atmospheric Parameter Air Temperature		Day -5	
Atmospheric Parameter Relative Humidity			Day -5
Atmospheric Parameter OLR		Day -3	
Atmospheric Parameter Surface Temperature		Day -4	

Table 5.2. The Anomaly Observation with days of selected parameters.

6 CONCLUSIONS

Atmospheric indices variations and ionospheric anomalies in data of Haiti earth quake M_w 7.2 on 14^{th} August 2021, has been analyzed from different ground instruments data and space vehicles data All results showing anomalies, thus confirming the Lithosphere Atmosphere Ionosphere Coupling phenomenon during the seismic preparation period of this earth quake as epicenter is located on complex fault zones system and at the boundary of two tectonic plates. The findings in the study are summarized as below:

Atmospheric parameters have shown maximum deviation within 5-day time window before main shock day of Haiti earth quake, as data indicated maximum deviation from upper bound for Air

Temperature time series values and maximum deviation from lower bound for Relative Humidity time series values on day -5. Rise in Outgoing Longwave Radiation (OLR) values has been observed on day -3 but on other hand maximum value of Surface Temperature (ST) has been observed on day -4. Atmospheric data has indicated variations which are well beyond predefined upper and lower bound limits in 5 days window prior to Haiti EQ 2021

The ionospheric anomalies have been indicated in data. Positive enhancement in VTEC values beyond the upper bound during the day time has been observed within 6-day time window before main shock day in three out of four IGS located in seismic region. The TEC anomalies from GNSS prominently crossed the upper bound limit by an amount of 10-12 TECU during UT=08-12 hours. VTEC data from IGS located outside seismic region, in Mexico indicated anomalies on very small scale.

The two out of three Swarm satellites confirmed rise in plasma density and STEC on day -5, within 6-day time window which supports evidence of pre-seismic perturbations, before the main shock day. These pre-seismic ionospheric anomalies in Swarm satellites data also endorse the LAIC hypothesis before the main shock. The variations started to form over the epicenter in Dobrovolsky [24] zone on August 10, 2021 and reached to maximum enhancement on same day.

This study also confirms the previous finding of release of gases from epicentral cavities and its propagation to ionosphere. The observed anomalies from analysis of atmospheric and ionospheric data from seismic preparation period or from post-earth quake period, may be used in earth quake forecasting especially for areas located on complex fault zones system or located at the boundary of two tectonic plates.

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